

# Hierarchical Acoustic-Semantic Modeling: Modality Separation and Semantic Coherence for Full-Duplex SLMs

## 分层声学-语义建模：面向全双工 SLMs 的模态分离与语义一致性

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<https://huggingface.co/HIT-TMG/Lychee-FD>

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## Abstract

### 摘要

Developing seamless, high-performance, native intelligent full-duplex Spoken Language Models (SLMs) remains a critical challenge and long-standing goal for the speech and NLP community. Despite notable progress, recent endeavors are fundamentally constrained by severe modality interference, which causes substantial knowledge degradation and compromises semantic integrity — ultimately making full-duplex SLMs feel unnatural and unintelligent. In this paper, through an exhaustive fine-grained analysis of model optimization dynamics, we uncover the root cause of such performance degradation, revealing that modality interference arises from inherent gradient conflicts between acoustic and semantic modeling when the two modalities are forced to share a deep parameter space. Guided by this key insight, we introduce Lychee-FD, a na-

tive end-to-end full-duplex framework designed to mitigate modality interference. Importantly, we propose a hierarchical parameter separation strategy that decouples conflicting modalities in deep layers while preserving cross-modality coherence via a dedicated semantic alignment channel. Extensive experiments on multiple full-duplex benchmarks demonstrate that our method significantly advances the state of the art, yielding substantial improvements in both speech intelligence (+7.4% on Spoken QA) and full-duplex interaction fluidity (+28.5% on FullDuplexBench 1.5) without compromising inference efficiency. To the best of our knowledge, this work is the first to achieve two key advances: 1) uncovering and elucidating the root cause of modality interference in full-duplex SLMs, and 2) designing an elegant hierarchical model together with a practical solution for seamless, high-performance, native intelligent full-duplex SLMs.

开发无缝、高性能、原生智能的全双工语音语言模型（SLMs）仍然是语音和自然语言处理（NLP）社区的一个关键挑战和长期目标。尽管取得了显著进展，但最近的努力在根本上受到严重模态干扰的限制，这导致了知识退化并损害了语义完整性——最终使全双工 SLMs 显得不自然和不智能。在本文中，通过详尽的细粒度模型优化动态分析，我们揭示了这种性能退化的根本原因，发现模态干扰源于在两个模态被迫共享深层参数空间时，声学建模和语义建模之间固有的梯度冲突。基于这一关键洞察，我们引入了 Lychee-FD，一个原生端到端的全双工框架，旨在缓解模态干扰。重要的是，我们提出了一种分层参数分离策略，该策略在深层网络中分离冲突的模态，同时通过专用的语义对齐通道保持跨模态的一致性。在多个全双工基准测试上的广泛实验表明，我们的方法显著提升了技术水平，在语音智能（在 Spoken QA 上提升 7.4%）和全双工交互流畅性（在 FullDuplexBench 1.5 上提升 28.5%）方面均取得了显著提升，而不会牺牲推理效率。据我们所知，这项工作首次实现两个关键进展：1) 揭示并阐明全双工 SLMs 中模态干扰的根本原因；2) 设计了一个优雅的分层模型以及一种实用的解决方案，以实现无缝、高性能、原生智能的全双工 SLMs。

## 1 Introduction

### 1 引言

The rapid evolution of Large Language Models (LLMs) has fundamentally reshaped our daily lives, establishing them as ubiquitous assistants capable of complex reasoning and instruction following. Within this landscape, Spoken Language Models (SLMs) represent a significant paradigm shift from text-based to voice-based interaction. Despite recent advancements in Omni-modal models capable of seamless voice interaction (Hurst et al., 2024; Zhan et al., 2024; Li et al., 2025b; Kaplan et al., 2020; Hoffmann et al., 2022; Xu et al., 2023;

大型语言模型（LLMs）的快速发展从根本上改变了我们的日常生活，使它们成为能够进行复杂推理和遵循指令的普遍助手。在这一背景下，语音语言模型（SLMs）代表了一种从基于文本的交互向基于语音的交互的重要范式转变。尽管在能够实现无缝语音交互的多功能模型方面取得了近期进展（Hurst 等，2024；Zhan 等，2024；Li 等，2025b；Kaplan 等，2020；Hoffmann 等，2022；Xu 等，2023；

Gao et al., 2025), a critical disparity remains between artificial agents and authentic human conversation. Currently, most voice interactions are constrained to a rigid half-duplex mode, where the system strictly alternates between listening and speaking in a sequential manner. In contrast, authentic human conversation is inherently full-duplex, requiring the ability to continuously process incoming audio streams while concurrently generating responses (Sacks et al., 1974; Fu et al., 2025; Ji et al., 2024). The forced turn-taking of half-duplex systems disrupts the fluidity of interaction, creating an artificial barrier between user and agent.

(Gao 等，2025)，人工代理与真实人类对话之间仍存在关键差异。目前，大多数语音交互都受限于一中严格的半双工模式，系统严格按顺序交替进行听和说。相比之下，真实的人类对话本质上是全双工的，需要能够连续处理传入的音频流，同时生成响应（Sacks 等，1974；Fu 等，2025；Ji 等，2024）。半双工系统的强制轮流机制会破坏交互的流畅性，在用户与代理之间造成人为的障碍。

To bridge this gap, developing native Full-Duplex SLMs (FDSLMS) has emerged as a critical challenge and a long-standing goal for the speech and NLP communities (Sacks et al., 1974; Arora et al., 2025; Nguyen et al., 2023; Lin et al., 2022). This full-duplex paradigm demands complex pragmatic reasoning, enabling the agent to handle spontaneous interruptions, inject natural backchannels,

and dynamically manage turn-taking without explicit system-level triggers (Hu et al., 2025; Liu et al., 2025). Consequently, achieving seamless full-duplex communication is widely regarded as the next critical milestone in human-machine interaction, promising to unlock a truly natural, fluid, and immersive conversation experience (Lin et al., 2025b; Fu et al., 2025; Lin et al., 2025a).

为了解决这一差距，开发原生的全双工说话人语言模型 (FDSLMS) 已成为语音和自然语言处理 (NLP) 社区面临的关键挑战和长期目标 (Sacks 等, 1974; Arora 等, 2025; Nguyen 等, 2023; Lin 等, 2022)。这种全双工范式需要复杂的语用推理，使代理能够处理自发的中断、注入自然的反馈信号，并在没有显式系统级触发的情况下动态管理对话轮次 (Hu 等, 2025; Liu 等, 2025)。因此，实现无缝的全双工通信被广泛认为是人机交互的下一个关键里程碑，有望解锁真正自然、流畅且沉浸式的对话体验 (Lin 等, 2025b; Fu 等, 2025; Lin 等, 2025a)。

Despite progress in FDSLMS development (Fu et al., 2024; Zeng et al., 2025), current methods still suffer from severe Modality Interference. As illustrated in Figure 1, adapting a half-duplex base model (black star) into a native End-to-End architecture (blue diamond) precipitates significant knowledge degradation. This degradation is further corroborated by state-of-the-art models such as Moshi (Défossez et al., 2024), which reports a significant drop in speech intelligence (-12.7% on LlamaQ) after full-duplex alignment. Historically, this type of interference has been a notorious and persistent optimization bottleneck across deep learning (Ruder, 2017; Caruana, 1997; Yu et al., 2020;

尽管在 FDSLMS 开发方面取得了进展 (Fu 等, 2024; Zeng 等, 2025)，当前的方法仍然受到严重的模态干扰的影响。如图 1 所示，将一个半双工基础模型（黑色星形）适应为原生的端到端架构（蓝色菱形）会导致显著的知识退化。这种退化在诸如 Moshi (Défossez 等, 2024) 等最先进的模型中也得到了进一步证实，该模型在完全双工对齐后报告了语音智能的显著下降（在 LlamaQ 上下降了 -12.7%）。历史上，这种类型的干扰一直是深度学习领域臭名昭著且顽固的优化瓶颈 (Ruder, 2017; Caruana, 1997; Yu 等, 2020;

Sener & Koltun, 2018). While some approaches (Wang et al., 2025; Chen et al., 2025b) attempt to circumvent this interference by adopting Thinker-Talker architectures (red diamond), they require intricate multi-stage training and introduce significant latency. These limitations indicate that existing paradigms either fail in knowledge retention or inference efficiency. Consequently, the central research question of this work is: How can we resolve this modality interference to simultaneously achieve high inference efficiency and robust knowledge retention in full-duplex SLMs?

(Sener & Koltun, 2018)。虽然一些方法 (Wang 等人, 2025; Chen 等人, 2025b) 尝试通过采用 Thinker-Talker 架构（红色菱形）来规避这种干扰，但它们需要复杂的多阶段训练，并引入了显著的延迟。这些限制表明，现有的范式要么在知识保留上失败，要么在推理效率上表现不佳。因此，本工作的核心研究问题是：我们如何解决这种模态干扰，以在全双工 SLMs 中同时实现高推理效率和稳健的知识保留？

To uncover the root cause of modality interference, we conduct an exhaustive fine-grained model optimization dynamics analysis (shown in Figure 2). By quantifying the geometric relationships between the gradient vectors of semantic and acoustic objectives, we demonstrate the inherent gradient conflicts that fundamentally cause this interference. First, by calculating the layer-wise gradient cosine similarity, we reveal severe optimization divergence in gradient directions. While the gradient directions of text and speech objectives are synergistic in shallow layers, they become increasingly orthogonal and negative in deeper layers. This empirically demonstrates that forcing acoustic and semantic modeling to update within a shared parameter space inevitably fractures the optimization trajectory. Second, by evaluating the gradient magnitude ratio, we identify severe semantic dilution. The temporal alignment of sparse text tokens (e.g., 3Hz) with dense audio frames (e.g., 25Hz) via padding tokens consistently suppresses the magnitude of semantic gradients across all layers. Consequently, the optimization landscape becomes overwhelmingly dominated by acoustic modeling, effectively suppressing semantic modeling.

为揭示模态干扰的根本原因，我们进行了全面的细粒度模型优化动态分析（如图 2 所示）。通过量化语义和声学目标梯度向量之间的几何关系，我们展示了导致这种干扰的固有梯度冲突。首先，通过计算层间梯度余弦相似度，我们揭示了梯度方向上的严重优化发散。在浅层中，文本和语音目标的梯度方向是协同的，但在深层中，它们逐渐变得正交且为负。这实证地表明，迫使声学和语义建模在共享参数空间中更新，不

可避免地会破坏优化轨迹。其次，通过评估梯度幅度比值，我们识别出严重的语义稀释现象。通过填充标记将稀疏文本标记（例如 3Hz）与密集音频帧（例如 25Hz）进行时序对齐，持续抑制了所有层中语义梯度的幅度。因此，优化景观被声学建模所主导，有效地抑制了语义建模。

Inspired by these insights, we introduce Lychee-FD, a native end-to-end full-duplex framework designed to mitigate modality interference with two architectural innovations. First, we propose a hierarchical parameter separation strategy for optimization divergence in deep layers. Specifically, we separate the deep layers into independent acoustic and semantic heads. By executing these heads in parallel, we maintain the original model depth, thereby preserving inference efficiency. Second, to counter semantic dilution, we introduce

受这些见解的启发，我们引入了 Lychee-FD，这是一个原生的端到端全双工框架，旨在通过两项架构创新来缓解模态干扰。首先，我们提出了一种分层参数分离策略，用于深度层中的优化发散。具体来说，我们将深度层分离为独立的声学 and 语义头。通过并行执行这些头，我们保持了原始模型的深度，从而保留了推理效率。其次，为了应对语义稀释问题，我们引入了

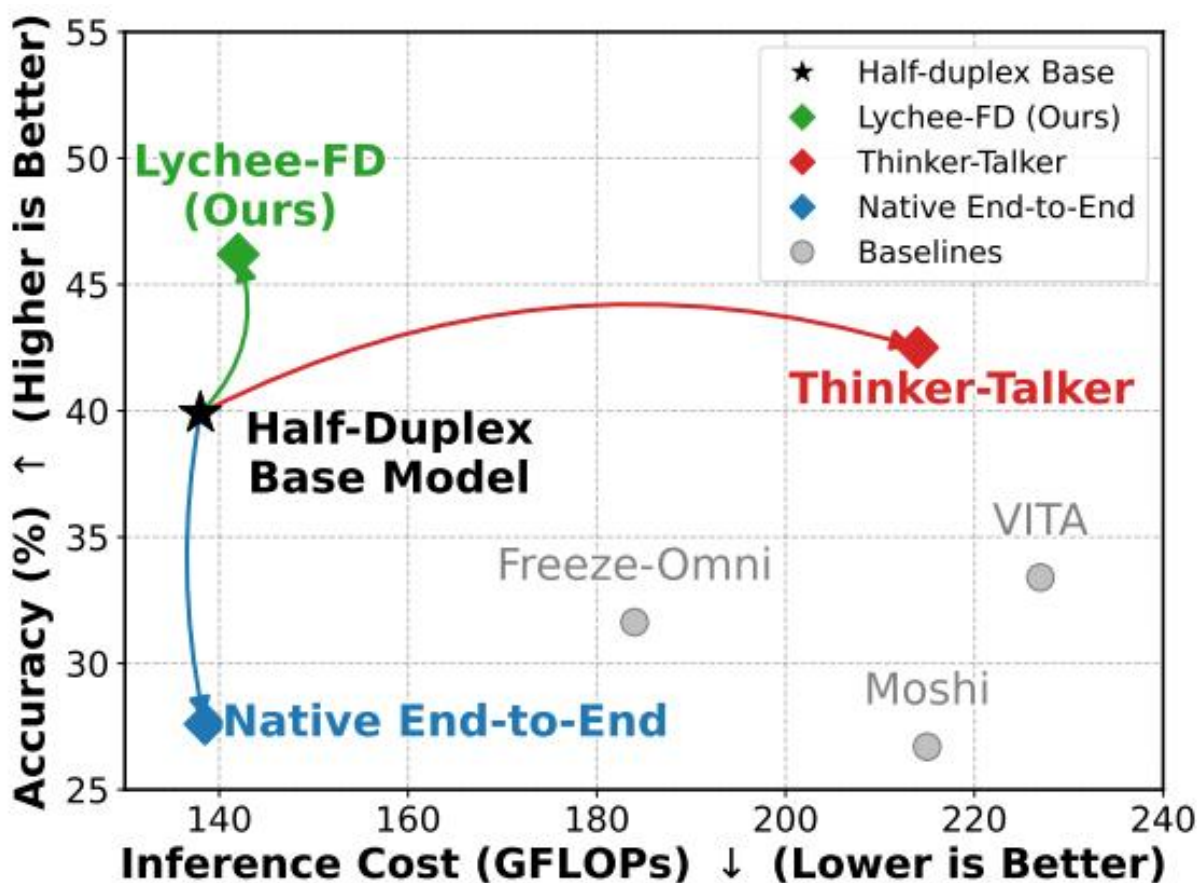


Figure 1: images/1515599a82c638e1f869e3660d90d36098f5efde4211f9e2dd98eadded91bc67.jpg

Figure 1: Visualization of the efficiency and intelligence trade-off. Existing paradigms face a dilemma when extending a half-duplex SLM (black star) to a full-duplex one. Native End-to-End models (blue diamond) sacrifice accuracy for efficiency, while Thinker-Talker models (red diamond) preserve knowledge but incur prohibitive inference costs. In contrast, our proposed Hierarchical framework (green diamond) combines low latency with high accuracy, significantly outperforming all full-duplex SLM baselines.

图 1：效率与智能性的权衡可视化。现有范式在将半双工 SLM（黑色星形）扩展为全双工 SLM 时面临困境。原生端到端模型（蓝色菱形）为了提高效率而牺牲了准确性，而 Thinker-Talker 模型（红色菱形）则保留了知识，但导致推理成本过高。相比之下，我们提出的分层框架（绿色菱形）结合了低延迟与高准确

性，在所有全双工 SLM 基线模型中表现显著优于其他模型。

a semantic alignment channel to generate coherent internal monologues. By utilizing continuous textual supervision as a semantic anchor, we preserve the robustness of semantic modeling during training. Extensive experiments demonstrate that Lychee-FD achieves the state-of-the-art performance, specifically delivering an average 7.4% improvement on Spoken QA tasks and a 28.5% gain on FullDuplexBench 1.5. Ultimately, our approach effectively mitigates modality interference, simultaneously achieving high inference efficiency and robust knowledge retention.

一个语义对齐通道，以生成连贯的内心独白。通过利用连续文本监督作为语义锚点，我们在训练过程中保持了语义建模的鲁棒性。大量实验表明，Lychee-FD 达到了最先进的性能，特别是在口语问答任务上平均提升了 7.4%，在 FullDuplexBench 1.5 上提升了 28.5%。最终，我们的方法有效缓解了模态干扰，同时实现了高推理效率和强大的知识保留。

Our contributions are summarized as follows:

我们的贡献总结如下：

- For the first time, we uncover the root cause of modality interference in full-duplex SLMs. Supported by an exhaustive fine-grained analysis of model optimization dynamics, we reveal that these obstacles stem from inherent gradient conflicts between acoustic and semantic modeling when forced into a shared deep parameter space.
- We present Lychee-FD, the first fully native intelligent full-duplex framework. We propose a hierarchical parameter separation strategy that decouples acoustic and semantic modeling, bridged by an elegantly designed semantic alignment channel to preserve coherent internal monologues and robust semantics.
- We advance the state-of-the-art across multiple full-duplex benchmarks. To forge ahead the full-duplex research in the communities, we open-source our framework, training pipeline, and model weights at <https://github.com/HITSz-TMG/Lychee-FD>.
- 首次，我们揭示了全双工 SLMs 中模态干扰的根本原因。通过详尽的细粒度模型优化动态分析，我们发现这些障碍源于在强制共享的深层参数空间中，声学建模与语义建模之间固有的梯度冲突。
- 我们提出了 Lychee-FD，这是首个完全原生的智能全双工框架。我们提出了一种分层参数分离策略，通过精心设计的语义对齐通道，将声学建模与语义建模解耦，以保持连贯的内部独白和稳健的语义。
- 我们在多个全双工基准测试中提升了当前的最先进水平。为了推动全双工研究的发展，我们在 <https://github.com/HITSz-TMG/Lychee-FD> 开源了我们的框架、训练流程和模型权重。

## 2 Related Work

## 2 相关工作

### 2.1 Spoken Language Model

#### 2.1 口语语言模型

SLMs have evolved from cascading ASR-LLM-TTS pipelines (Zhang et al., 2023; An et al., 2024; Borsos et al., 2023; Chen et al., 2025c; Zeghidour et al., 2022) to unified architectures. Current methods are mainly grouped into two paradigms:

SLMs 已从级联的 ASR-LLM-TTS 管道 (Zhang 等, 2023; An 等, 2024; Borsos 等, 2023; Chen 等, 2025c; Zeghidour 等, 2022) 演进为统一的架构。当前的方法主要分为两种范式:

**Thinker-Talker Architectures.** This paradigm decouples acoustic generation from the LLM backbone to mitigate modality interference (Xu et al., 2025; Wang et al., 2025; Fang et al., 2025). For instance, Xu et al. (2025) adopts a dual-track autoregressive architecture. Despite their stability, these systems often require intricate, multi-stage training curricula and suffer from inference bottlenecks caused by the separate generation modules.

思考者-交谈者架构。这种范式将声学生成与 LLM 主干解耦, 以减轻模态干扰 (Xu 等, 2025; Wang 等, 2025; Fang 等, 2025)。例如, Xu 等 (2025) 采用了一种双轨自回归架构。尽管这些系统具有稳定性, 但它们通常需要复杂、多阶段的训练课程, 并且由于独立生成模块而受到推理瓶颈的困扰。

**Native End-to-End Architectures.** Conversely, these architectures embed speech and text into a shared semantic space, enabling direct speech-to-speech reasoning (Li et al., 2025c; Xie & Wu, 2024a; Mitsui et al., 2024; Zeng et al., 2025). For examples, Wu et al. (2025) introduces latent audio encoding to capture paralinguistic cues.

原生的端到端架构。相反, 这些架构将语音和文本嵌入到一个共享的语义空间中, 从而实现直接的语音到语音推理 (Li 等, 2025c; Xie & Wu, 2024a; Mitsui 等, 2024; Zeng 等, 2025)。例如, Wu 等 (2025) 引入了潜在的音频编码以捕捉副语言线索。

Crucially, while existing SLMs remain fundamentally bottlenecked by rigid, half-duplex turn-taking mechanisms, our work advances the field by establishing a native Full-Duplex framework. By enabling simultaneous listening and speaking, we aim to unlock a truly natural, fluid, and immersive paradigm for future human-machine interaction.

关键的是, 尽管现有的大语言模型 (SLMs) 仍然受到刚性、半双工轮流机制的根本性限制, 我们的工作通过建立原生的全双工框架, 推动了该领域的发展。通过实现同时听和说, 我们旨在为未来的人机交互解锁一种真正自然、流畅且沉浸的范式。

## 2.2 Full Duplex Speech Interaction

### 2.2 全双工语音交互

To achieve turn-free and fluid conversation, recent research has explored full-duplex paradigms that transcend rigid turn-taking. These efforts can be broadly categorized into three paradigms:

为了实现无回合、流畅的对话, 近期的研究探索了超越严格回合制的全双工范式。这些努力可以大致分为以下三种范式:

**Full-Duplex Dialogue System.** Early efforts primarily leverage Voice Activity Detection (VAD) as a dialogue manager to control the speaking process of half-duplex SLMs (Chen et al., 2025a; Liao et al., 2025; Li et al., 2025a; Zhang et al., 2025a). Although making some progress, these cascaded systems suffer from high latency and error propagation, prompting a shift towards unified modeling.

全双工对话系统。早期的研究主要利用语音活动检测 (VAD) 作为对话管理器, 以控制半双工 SLM (Chen et al., 2025a; Liao et al., 2025; Li et al., 2025a; Zhang et al., 2025a) 的说话过程。尽管取得了一定进展, 这些级联系统仍存在高延迟和误差传播的问题, 促使研究转向统一建模。

**Time-Division Multiplexing (TDM).** To address these limitations, recent works use the SLM itself as the dialogue manager. TDM approaches flatten listening and speaking tokens into a single temporal sequence. However, this leads to increasing computational complexity and limits long-context interaction (Zhang et al., 2025b; Veluri et al., 2024; Zhang et al., 2024; Yu et al., 2025).

时间分割复用 (Time-Division Multiplexing, TDM)。为了解决这些限制, 近期的研究工作将 SLM 本身

作为对话管理器。TDM 方法将听和说的标记扁平化为一个单一的时间序列。然而，这会导致计算复杂度的增加，并限制了长上下文交互（Zhang 等，2025b；Veluri 等，2024；Zhang 等，2024；Yu 等，2025）。

Channel-Division Multiplexing (CDM). Instead of flattening, CDM approaches explicitly model concurrent input and output streams, offering the most integrated form of interaction (Zhang et al., 2025c; Nguyen et al., 2023; Xie & Wu, 2024b). For example, Défossez et al. (2024) and Chen et al. (2025b) integrate time-aligned text and audio streams to provide explicit semantic guidance for speech generation. However, these fully shared architectures entangle distinct modalities in a single parameter space, leaving them highly susceptible to severe knowledge degradation.

频道分割复用（CDM）。与扁平化方法不同，CDM 方法明确建模并发的输入和输出流，提供了最集成的交互形式（Zhang 等，2025c；Nguyen 等，2023；Xie & Wu，2024b）。例如，Défossez 等（2024）和 Chen 等（2025b）将时间对齐的文本和音频流集成起来，为语音生成提供显式的语义引导。然而，这些完全共享的架构将在单一参数空间中纠缠不同的模态，使它们极易受到严重知识退化的困扰。

Unlike previous works that struggle with modality interference, Lychee-FD provides the first fundamental solution to the core problem within FDSLMS. By resolving the inherent deep-layer gradient conflicts between acoustic and semantic modeling, our native end-to-end framework successfully achieves ultra-low latency with robust knowledge retention, substantially pushing forward the frontier of research for the speech and NLP communities.

与以往难以处理模态干扰的研究所不同，Lychee-FD 为 FDSLMS 中的核心问题提供了首个根本性的解决方案。通过解决声学建模与语义建模之间固有的深层梯度冲突，我们的原生端到端框架成功实现了超低延迟并保持稳健的知识保留，显著推动了语音和 NLP 社区的研究前沿。

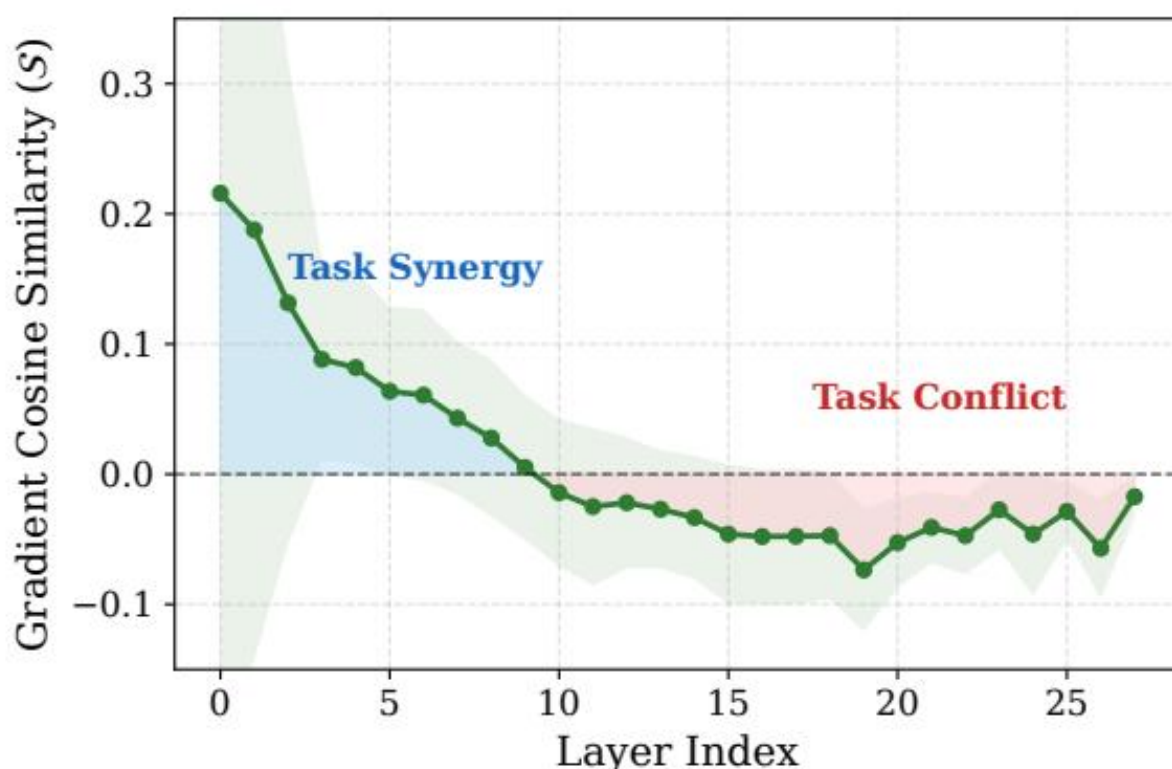


Figure 2: images/e1c0905ea5dd586dde139e646f5358e2fe6837f2c527101188d156c265bbd0ad.jpg

(a) Gradient Cosine Similarity

(a) 梯度余弦相似度



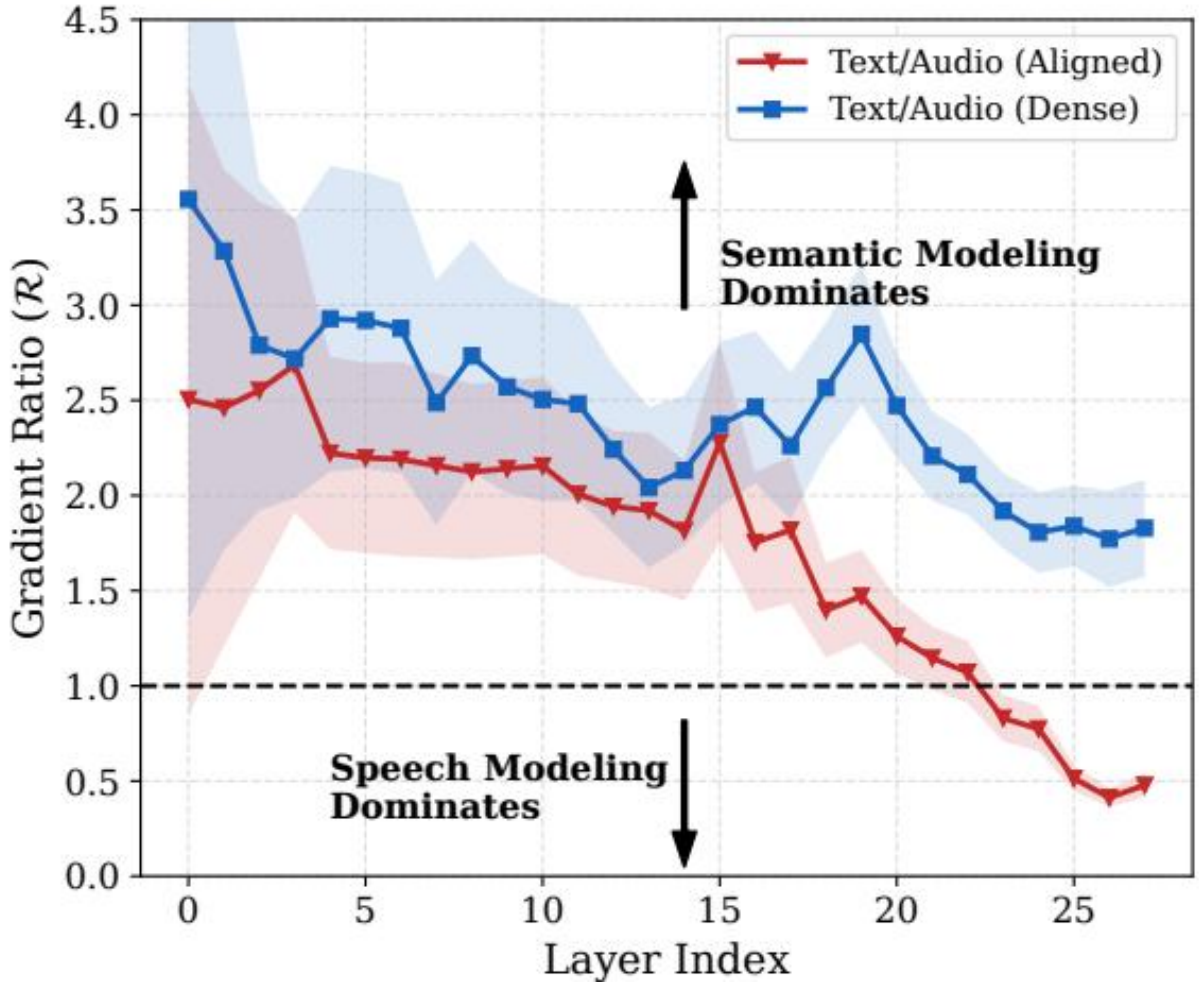


Figure 3: images/8e91b4110c7c99a6be3d751bac7b6b7e7513ea321d46adedec9de22d262beadd.jpg

(b) Gradient Magnitude Ratio

Figure 2: Optimization Dynamics Visualization. (a) Gradient Cosine Similarity: The transition to negative values in deep layers reveals conflicting optimization directions between semantic and acoustic modeling, motivating our hierarchical parameter separation. (b) Gradient Magnitude Ratio: The consistently lower ratio for “Aligned” (Red) compared to “Dense” (Blue) indicates that sparse alignment dilutes semantic supervision, motivating our semantic alignment channel.

(b) 梯度幅值比

图 2：优化动态可视化。(a) 梯度余弦相似度：在深层网络中向负值的转变揭示了语义建模和声学建模之间优化方向的冲突，这促使我们提出分层参数分离的方法。(b) 梯度幅值比：与“Dense”（蓝色）相比，“Aligned”（红色）始终具有较低的比率，这表明稀疏对齐稀释了语义监督，这促使我们提出语义对齐通道。



## 3 Hierarchical Acoustic-Semantic Modeling

### 3 分层的声学-语义建模

#### 3.1 Model Optimization Dynamics Analysis of Modality Interference

##### 3.1 模态干扰的模型优化动态分析

To empirically investigate the fundamental root cause of the modality interference and understand the learning process within FDSLMS, we conducted an in-depth model optimization dynamics analysis, shown in Figure 2. We utilized a native CDM architecture initialized with weights from StepAudio-2-mini (Wu et al., 2025). We performed forward passes on 1K samples from the training set to accumulate gradients for the cross-entropy losses of both text token generation ( $L_{text}$ ) and speech token generation ( $L_{speech}$ ), without updating the model parameters. Specifically, the layer-wise gradient vectors are formally defined as follows:

为了实证研究模态干扰的根本原因并理解 FDSLMS 中的学习过程，我们进行了深入的模型优化动态分析，如图 2 所示。我们使用了一个以 StepAudio-2-mini (Wu 等, 2025) 权重初始化的原生 CDM 架构。我们在训练集的 1K 个样本上进行前向传递，以累积文本标记生成 ( $L_{text}$ ) 和语音标记生成 ( $L_{speech}$ ) 的交叉熵损失的梯度，而无需更新模型参数。具体而言，逐层梯度向量的正式定义如下：

$$\mathbf{g}_{\text{text}}^{(l)} = \nabla_{\theta^{(l)}} \mathcal{L}_{\text{text}}, \quad (1)$$

$$\mathbf{g}_{\text{speech}}^{(l)} = \nabla_{\theta^{(l)}} \mathcal{L}_{\text{speech}}, \quad (2)$$

where  $\nabla_{\theta^{(l)}}$  denotes the gradient operator with respect to the layer parameters  $\theta^{(l)}$ , and the resulting tensors are flattened into vectors. By analyzing the geometric relationships of these gradient vectors, we quantitatively uncover the fundamental interaction dynamics between the two modalities during the entire learning process.

其中  $\nabla_{\theta^{(l)}}$  表示相对于层参数  $\theta^{(l)}$  的梯度算子，所得张量被展平为向量。通过分析这些梯度向量的几何关系，我们定量地揭示了两种模态在整个学习过程中基本的相互作用动态。

**Optimization Divergence.** We investigated the compatibility of the two optimization objectives by calculating the cosine similarity  $\mathcal{S}^{(l)}$  between text and speech gradient vectors:

优化发散。我们通过计算文本和语音梯度向量之间的余弦相似度  $\mathcal{S}^{(l)}$ ，来研究两个优化目标的兼容性：

$$\mathcal{S}^{(l)} = \cos(\mathbf{g}_{\text{text}}^{(l)}, \mathbf{g}_{\text{speech}}^{(l)}). \quad (3)$$

As shown in Figure 2a, the similarity reveals a distinct layer-wise pattern. In the shallow layers (0-9), the cosine similarity is positive, indicating that the two modalities share synergistic optimization directions, focusing on common low-level features processing. However, as the depth increases, the similarity drops sharply, turning negative and fluctuating in the deeper layers. This trend empirically confirms our hypothesis regarding the dual nature of speech: while shallow layers can share representations, the deep layers face a fundamental conflict between acoustic and semantic modeling. Forcing a unified set of parameters to resolve these opposing gradient directions inevitably leads to sub-optimal performance, validating the root cause of the observed modality interference. Moreover, we confirm these findings on Moshi architecture in Section

5 and provide a global gradient influence analysis in in Appendix 9 to demonstrate the causal destructive impact of this modality conflict.

如图 2a 所示，相似性揭示了一个明显的分层模式。在浅层（0-9）中，余弦相似性为正值，表明两种模态具有协同优化方向，专注于共同的低级特征处理。然而，随着深度的增加，相似性急剧下降，并在更深的层中变为负值并波动。这一趋势实证验证了我们关于语音双重性质的假设：虽然浅层可以共享表示，但深层面临声学建模与语义建模之间的根本性冲突。强制使用统一的参数集来解决这些相反的梯度方向不可避免地导致次优性能，从而验证了观察到的模态干扰的根本原因。此外，在第 5 节中我们确认了这些发现，并在附录 9 中提供了全局梯度影响分析，以展示这种模态冲突的因果破坏性影响。

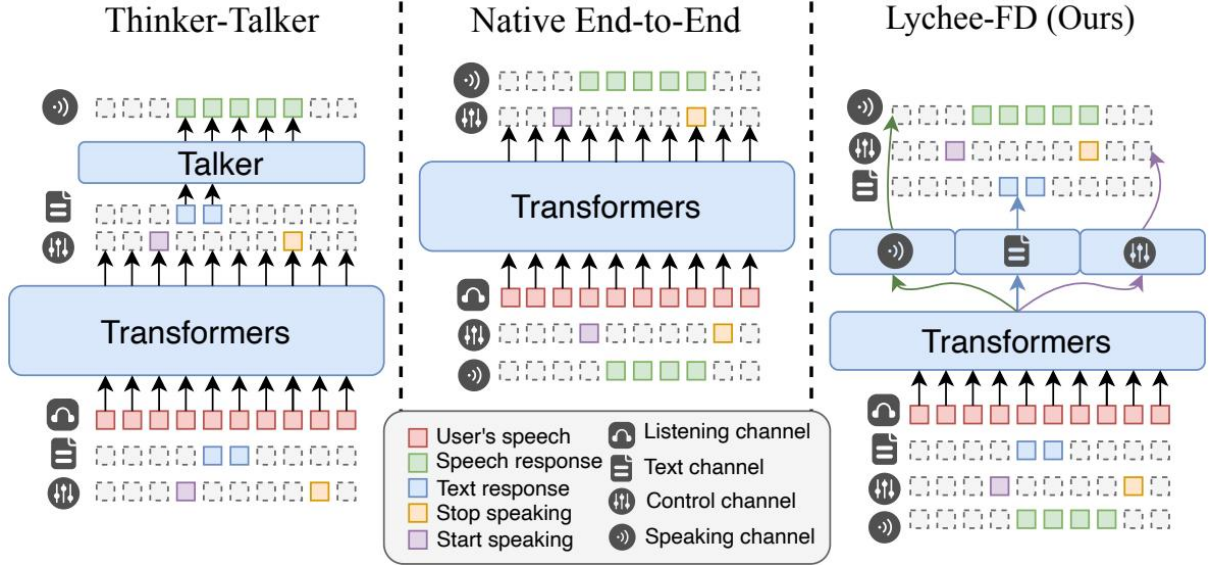


Figure 4: images/ecfbb6fe7b3928cb46586dd5dd73e04ddcf2da1eda28111917fb9517558b1bcd.jpg

Figure 3: Two mainstream architecture paradigms of SLMs and our proposed Lychee-FD. Our design features a hierarchical parameter separation strategy to resolve deep-layer modality conflicts and a semantic alignment channel to enforce robust knowledge retention without sacrificing inference efficiency.

图 3：SLMs 的两种主流架构范式以及我们提出的 Lychee-FD。我们的设计采用分层参数分离策略以解决深层模态冲突，并通过语义对齐通道来确保在不牺牲推理效率的前提下实现鲁棒的知识保留。

**Semantic Dilution.** Prevalent Full-Duplex SLMs typically address the frequency mismatch between text (approximately 3Hz) and audio (typically 25Hz) by interleaving padding tokens to enforce temporal alignment (Défossez et al., 2024; Wu et al., 2025; Chen et al., 2025b). To evaluate the impact of this alignment on optimization, we compared the ratio  $\mathcal{R}^{(l)}$  of gradient magnitudes between the two modalities:

语义稀释。常见的全双工 SLMs 通常通过插入填充标记来强制时间对齐，以解决文本（大约 3Hz）和音频（通常为 25Hz）之间的频率不匹配问题（Défossez 等，2024；Wu 等，2025；Chen 等，2025b）。为了评估这种对齐对优化的影响，我们比较了两种模态之间梯度幅度的比率  $\mathcal{R}^{(l)}$ ：

$$\mathcal{R}^{(l)} = \frac{\|\mathbf{g}_{\text{text}}^{(l)}\|}{\|\mathbf{g}_{\text{speech}}^{(l)}\|}. \quad (4)$$

As illustrated in Figure 2b, we observe a substantial disparity between the continuous text supervision (Dense) and the sparse text with padding (Aligned). Specifically, the gradient magnitude ratio in the Aligned setting is consistently suppressed across all layers, suggesting that the introduction of padding tokens effectively dilutes the density of semantic supervision. Consequently,

the optimization dynamics become dominated by acoustic reconstruction, which drives the observed degradation in knowledge retention.

如图 2b 所示，我们观察到连续文本监督 (Dense) 与带有填充的稀疏文本 (Aligned) 之间存在显著差异。具体而言，在 Aligned 设置中，梯度幅度比值在所有层中都持续受到抑制，这表明引入填充标记有效地稀释了语义监督的密度。因此，优化动态被主导于声学重建，这导致了知识保留能力下降所观察到的退化现象。

## 3.2 Model Design: Modality Separation and Semantic Coherence

### 3.2 模型设计：模态分离与语义一致性

Guided by these insights, we introduce Lychee-FD, a native end-to-end framework designed to resolve modality interference. Rather than circumventing these optimization bottlenecks via high-latency Thinker-Talker models (as illustrated in Figure 3), our architecture fundamentally tackles them via two targeted innovations: a Hierarchical Parameter Separation strategy to disentangle conflicting optimization directions, and a Semantic Alignment Channel to counteract semantic dilution and enforce robust knowledge retention.

基于这些见解，我们引入了 Lychee-FD，这是一个原生的端到端框架，旨在解决模态干扰问题。与通过高延迟的 Thinker-Talker 模型绕过这些优化瓶颈（如图 3 所示）不同，我们的架构通过两项针对性创新从根本上解决这些问题：一种分层参数分离策略，用于分离冲突的优化方向；以及一个语义对齐通道，用于对抗语义稀释并确保稳健的知识保留。

**Shared Backbone Foundation.** We choose StepAudio-2-mini (Wu et al., 2025) as our half-duplex backbone due to its public availability. We employ the Whisper-v3-large encoder for input audio processing. For acoustic output, we adopt the CosyVoice2 tokenizer to convert audio into discrete speech tokens at a 25Hz frame

共享骨干网络基础。我们选择 StepAudio-2-mini (Wu 等人, 2025) 作为我们的半双工骨干网络，因为其具有公共可用性。我们采用 Whisper-v3-large 编码器进行输入音频处理。对于声学输出，我们采用 CosyVoice2 分词器，将音频转换为离散的语音标记，采样率为 25Hz。

| Model                                                      | Type                                                       | LlamaQ                                                     |                                                            | WebQ                                                       |                                                            | TriviaQA                                                   |                                                            | Avg. |      | Takeover Rate |
|------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------|------|------|---------------|
| $\begin{smallmatrix} S \rightarrow \\ T \end{smallmatrix}$ | $\begin{smallmatrix} S \rightarrow \\ S \end{smallmatrix}$ | $\begin{smallmatrix} S \rightarrow \\ T \end{smallmatrix}$ | $\begin{smallmatrix} S \rightarrow \\ S \end{smallmatrix}$ | $\begin{smallmatrix} S \rightarrow \\ T \end{smallmatrix}$ | $\begin{smallmatrix} S \rightarrow \\ S \end{smallmatrix}$ | $\begin{smallmatrix} S \rightarrow \\ T \end{smallmatrix}$ | $\begin{smallmatrix} S \rightarrow \\ S \end{smallmatrix}$ |      |      |               |
| Freeze-Omni                                                | System-level                                               | 71.3                                                       | 50.7                                                       | 38.3                                                       | 25.8                                                       | 24.3                                                       | 23.9                                                       | 44.6 | 33.4 | 99.6          |
| VITA 1.5                                                   | System-level                                               | 75.7                                                       | 51.0                                                       | 41.8                                                       | 29.2                                                       | 35.0                                                       | 26.0                                                       | 50.8 | 35.4 | 100           |
| dGSLM                                                      | Native                                                     | -                                                          | 1.3                                                        | -                                                          | 0.2                                                        | -                                                          | 0.4                                                        | -    | 0.6  | 100           |
| FLM-audio                                                  | Native                                                     | 41.3                                                       | 36.7                                                       | 15.6                                                       | 14.5                                                       | 10.5                                                       | 10.4                                                       | 22.4 | 20.5 | 99.5          |
| Moshi                                                      | Native                                                     | 62.3                                                       | 54.7                                                       | 25.3                                                       | 19.6                                                       | 19.1                                                       | 17.4                                                       | 35.5 | 30.5 | 93.8          |
| SALMON omni*                                               | Native                                                     | 67.0                                                       | 61.7                                                       | 33.7                                                       | 28.1                                                       | 32.9                                                       | 24.2                                                       | 44.5 | 38.0 | 99.9          |
| Fun-Audio-Chat                                             | Native                                                     | 72.3                                                       | 64.3                                                       | 26.2                                                       | 24.4                                                       | 29.6                                                       | 27.7                                                       | 42.7 | 38.8 | 99.9          |
| StepAudio-2-mini                                           | Half-duplex                                                | 74.7                                                       | 62.0                                                       | 39.9                                                       | 30.8                                                       | 39.5                                                       | 29.8                                                       | 51.3 | 40.9 | -             |

|                  |        |      |      |      |      |      |      |      |      |      |
|------------------|--------|------|------|------|------|------|------|------|------|------|
| Lychee-FD (Ours) | Native | 73.7 | 65.3 | 38.3 | 33.9 | 42.5 | 39.4 | 51.5 | 46.2 | 100  |
| w/o Sem-Channel  |        | 69.3 | 61.0 | 34.1 | 31.5 | 34.2 | 30.1 | 45.9 | 40.8 | 99.6 |
| w/o Param-Sep    |        | 67.0 | 36.0 | 34.6 | 22.5 | 36.6 | 24.2 | 46.1 | 27.6 | 98.5 |

| 模型                | 类型                | LlamaQ            |                   | WebQ              |                   | TriviaQA          |                   | 平均值  |      | 接管率  |
|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|------|------|------|
| $S \rightarrow T$ | $S \rightarrow S$ | $S \rightarrow T$ | $S \rightarrow S$ | $S \rightarrow T$ | $S \rightarrow S$ | $S \rightarrow T$ | $S \rightarrow S$ |      |      |      |
| Freeze-Omni       | 系统级               | 71.3              | 50.7              | 38.3              | 25.8              | 24.3              | 23.9              | 44.6 | 33.4 | 99.6 |
| VITA 1.5          | 系统级               | 75.7              | 51.0              | 41.8              | 29.2              | 35.0              | 26.0              | 50.8 | 35.4 | 100  |
| dGSLM             | 原生                | -                 | 1.3               | -                 | 0.2               | -                 | 0.4               | -    | 0.6  | 100  |
| FLM-audio         | 原生                | 41.3              | 36.7              | 15.6              | 14.5              | 10.5              | 10.4              | 22.4 | 20.5 | 99.5 |
| 摩西                | 原生                | 62.3              | 54.7              | 25.3              | 19.6              | 19.1              | 17.4              | 35.5 | 30.5 | 93.8 |
| SALMON omni*      | 原生                | 67.0              | 61.7              | 33.7              | 28.1              | 32.9              | 24.2              | 44.5 | 38.0 | 99.9 |
| Fun-Audio-Chat    | 原生                | 72.3              | 64.3              | 26.2              | 24.4              | 29.6              | 27.7              | 42.7 | 38.8 | 99.9 |
| StepAudio2-mini   | 半双工               | 74.7              | 62.0              | 39.9              | 30.8              | 39.5              | 29.8              | 51.3 | 40.9 | -    |
| 荔枝-FD (我们的)       | 原生                | 73.7              | 65.3              | 38.3              | 33.9              | 42.5              | 39.4              | 51.5 | 46.2 | 100  |
| 无 Sem-Channel     |                   | 69.3              | 61.0              | 34.1              | 31.5              | 34.2              | 30.1              | 45.9 | 40.8 | 99.6 |
| 无参数分离             |                   | 67.0              | 36.0              | 34.6              | 22.5              | 36.6              | 24.2              | 46.1 | 27.6 | 98.5 |

Table 1: Performance comparison on spoken question answering benchmarks. We report accuracy (Acc) in both speech-to-text ( $S \rightarrow T$ ) and speech-to-speech ( $S \rightarrow S$ ) settings. We also report average takeover rate (TOR) across three benchmarks. Bold denotes best results and underlined denotes second best. \* denotes our implementation.

表 1: 在语音问答基准测试中的性能比较。我们在语音到文本 ( $S \rightarrow T$ ) 和语音到语音 ( $S \rightarrow S$ ) 设置中报告准确率 (Acc)。我们还报告了三个基准测试的平均接管率 (TOR)。粗体表示最佳结果，下划线表示次佳结果。\* 表示我们的实现。

rate. Crucially, to ensure precise temporal alignment for full-duplex interaction, we utilize a 25Hz frame rate, distinct from the setting adopted by Step-Audio-2.

速率。关键的是，为了确保全双工交互的精确时间对齐，我们采用 25Hz 的帧率，这与 Step-Audio-2 采用的设置不同。

**Hierarchical Parameter Separation.** Guided by the observation that acoustic and semantic gradients become orthogonal in deeper layers, we design a hierarchical Transformer architecture. We retain a unified Transformer backbone for the shallow layers to leverage shared low-level feature processing. Formally, given the input embedding sequence  $\mathbf{E} \in \mathbb{R}^{N \times d}$ , the shared representation  $\mathbf{H}_{\text{shared}}$  is mathematically formulated as follows:

层次化参数分离。基于观察到在更深层中，声学 and 语义梯度变得正交，我们设计了一种层次化的 Transformer 架构。我们保留一个统一的 Transformer 主干网络用于浅层，以利用共享的低级特征处理。正式地，给定输入嵌入序列  $\mathbf{E} \in \mathbb{R}^{N \times d}$ ，共享表示  $\mathbf{H}_{\text{shared}}$  的数学形式如下：

$$\mathbf{H}_{\text{shared}} = \mathcal{F}_{\text{shared}}(\mathbf{E}; \theta_{\text{shared}}), \quad (5)$$

where  $\mathcal{F}_{\text{shared}}$  denotes the stack of shared Transformer layers parameterized by  $\theta_{\text{shared}}$ .

其中  $\mathcal{F}_{\text{shared}}$  表示由  $\theta_{\text{shared}}$  参数化的共享 Transformer 层堆栈。

In the deeper layers, we physically disentangle the parameters into three specialized heads: the Semantic Head for text generation, the Acoustic Head for speech synthesis, and the Control Head for interaction management (e.g., stop/start signals). These heads operate in parallel as follows:

在更深层的结构中，我们物理上将参数拆分为三个专门的头部：用于文本生成的语义头部、用于语音合成的声学头部，以及用于交互管理的控制头部（例如，停止/启动信号）。这些头部并行运行，如下所示：

$$\mathbf{O}^m = \mathcal{F}_{\text{head}}^m(\mathbf{H}_{\text{shared}}; \theta_m), \quad m \in \{T, A, C\} \quad (6)$$

where  $m$  represents the modality (Text, Acoustic, Control), and  $\mathbf{O}^m$  denotes the output logits for each head. The total cross-entropy loss  $\mathcal{L}$  is computed as the summation of the next-token prediction losses for each specific head as follows:

其中  $m$  表示模态（文本、音频、控制）， $\mathbf{O}^m$  表示每个头的输出 logits。总的交叉熵损失  $\mathcal{L}$  计算为每个特定头的下一个 token 预测损失之和，如下所示：

$$\mathcal{L} = - \sum_{m \in \{T, A, C\}} \sum_t \log P(y_t^m | y_{<t}, \mathbf{E}; \theta), \quad (7)$$

where  $y_t^m$  denotes the ground-truth token for modality  $m$  at step  $t$ . This hierarchical split effectively isolates the conflicting optimization objectives, allowing the model to articulate high-fidelity acoustic responses without corrupting its underlying semantic modeling.

其中  $y_t^m$  表示第  $t$  步中模态  $m$  的真实标签。这种层次化的分割有效地隔离了冲突的优化目标，使模型能够在不破坏其底层语义建模的前提下，生成高保真的声学响应。

**Semantic Alignment Channel.** To counter the semantic dilution caused by sparse alignment in speech-native tasks, we incorporate a semantic alignment channel to generate coherent internal monologues. During training, these monologues serve as a semantic anchor, maintaining high-magnitude gradient flow for the language modeling objective. Specifically, we organize the parallel generation streams as follows:

语义对齐通道。为了应对语音原生任务中由于稀疏对齐导致的语义稀释问题，我们引入了一个语义对齐通道以生成连贯的内心独白。在训练过程中，这些独白作为语义锚点，保持语言建模目标的高幅度梯度流动。具体来说，我们按照以下方式组织并行生成流：

$$Y^T = [t_1, t_2, \dots, t_n, \langle \text{EOT} \rangle, \langle \text{pad} \rangle, \dots, \langle \text{pad} \rangle],$$

$$Y^A = [a_1, a_2, \dots, a_n, a_{n+1}, a_{n+2}, \dots, \langle \text{EOS} \rangle],$$

$$Y^C = [\langle \text{Start} \rangle, c_1, \dots, \dots, \dots, \langle \text{Stop} \rangle],$$

where  $Y^C$  employs special tokens (e.g., and ) to manage the onset and offset of the model's speech. By explicitly modeling the text channel alongside the acoustic channel, we ensure high-magnitude gradient flow for the language modeling objective, thereby preserving robust knowledge retention.

where  $Y^C$  使用特殊标记（例如和）来管理模型语音的开始和结束。通过显式地对文本通道和声学通道进行建模，我们确保语言建模目标下的高幅度梯度流动，从而保持稳健的知识保留。

## 4 Experiments

### 4 实验

#### 4.1 Baselines

##### 4.1 基线

To ensure a comprehensive evaluation, we compare our proposed method against a diverse set of representative and competitive full-duplex SLMs. These baselines cover the primary architectural paradigms currently explored in the field:

为确保全面评估，我们将所提出的算法与一系列具有代表性和竞争力的全双工 SLMs 进行比较。这些基线涵盖了目前该领域中主要探索的架构范式：

System-Level Full-Duplex Models include Freeze-Omni (Wang et al., 2025) and VITA-1.5 (Fu et al., 2025), which integrate an external VAD module to manage the dialogue state of a standard half-duplex SLM.

系统级全双工模型包括 Freeze-Omni (Wang 等人, 2025) 和 VITA-1.5 (Fu 等人, 2025)，这些模型集成了一个外部的 VAD 模块，用于管理标准半双工 SLM 的对话状态。

Native Full-Duplex Models include dGSLM (Nguyen et al., 2023), FLM-Audio (Yao et al., 2025), Moshi (Défossez et al., 2024) and Fun-Audio-Chat (Chen et al., 2025b). Specifically, Fun-Audio-Chat adopts a Thinker-Talker architecture, while the others utilize a CDM architecture.

原生全双工模型包括 dGSLM (Nguyen 等, 2023)、FLM-Audio (Yao 等, 2025)、Moshi (Défossez 等, 2024) 和 Fun-Audio-Chat (Chen 等, 2025b)。具体而言，Fun-Audio-Chat 采用的是 Thinker-Talker 架构，而其他模型则使用 CDM 架构。

#### 4.2 Training Data Construction

##### 4.2 训练数据构建

Given the scarcity of open-source full-duplex datasets, we developed an automated pipeline to synthesize high-quality training data covering three key interaction behaviors: Interruptions,



User Backchannels, and AI Backchannels. We employed multiple agents to simulate realistic User-Assistant dialogues, injecting rule-based constraints to trigger diverse interruption types (e.g., topic switching, follow-up queries) and natural backchannels. To ensure acoustic robustness, we synthesized speech from the transcripts via CosyVoice 2 Du et al. (2024), coupled with 80K predefined voice prompts for zero-shot cloning. After rigorous filtering to remove samples with logical inconsistencies or low audio quality, we curated a final dataset of approximately 140K full-duplex dialogue instances, providing a diverse and reliable foundation for our experiments. We provide more details of our data pipeline in Appendix 11.

鉴于开放源代码的全双工数据集稀缺，我们开发了一个自动化流程，合成高质量的训练数据，涵盖三种关键的交互行为：中断、用户回音和 AI 回音。我们使用多个代理来模拟真实的用户-助手对话，通过基于规则的约束触发多样化的中断类型（例如，话题切换、后续问题）和自然的回音。为确保音频鲁棒性，我们通过 CosyVoice 2 Du et al. (2024) 从转录文本合成语音，并结合 80,000 个预定义的语音提示进行零样本克隆。经过严格筛选以去除逻辑不一致或音频质量低的样本后，我们整理出一个最终包含约 14 万个全双工对话实例的数据集，为我们的实验提供了多样且可靠的基础。我们将在附录 11 中提供我们数据流程的更多细节。

## 4.3 Implementation Details

### 4.3 实施细节

We optimize our model using AdamW (Loshchilov & Hutter, 2019) with a cosine learning rate scheduler. All experiments are conducted on 8 NVIDIA H20 GPUs, with a global batch size of 32 and a learning rate of  $3e-6$ . We set the warmup ratio to 0.1 and train for 1 epoch, which takes approximately 16 hours. For inference, we employ greedy sampling for both text and speech token generation. We evaluate our model with three random seeds and report their average performance. Regarding the hierarchical architecture configuration, unless otherwise specified, we utilize a shared backbone of 24 Transformer layers. The specialized heads are configured with 4 layers for the text channel, 4 layers for the speech channel, and 2 layers for the control channel. This yields approximately 10B total parameters, a choice supported by the marginal gain analysis in Appendix 7.

我们使用 AdamW (Loshchilov & Hutter, 2019) 优化我们的模型，并采用余弦学习率调度器。所有实验均在 8 块 NVIDIA H20 GPU 上进行，全局批量大小为 32，学习率为  $3e-6$ 。我们将 warmup 比例设置为 0.1，并训练 1 个 epoch，这大约需要 16 小时。对于推理，我们对文本和语音 token 的生成均采用贪心采样。我们使用三个随机种子评估我们的模型，并报告其平均性能。关于层次化架构配置，除非另有说明，我们采用 24 层 Transformer 的共享主干。专用头部分别配置为文本通道 4 层、语音通道 4 层、控制通道 2 层。这使得总参数量约为 100 亿，这一选择得到了附录 7 中边际增益分析的支持。

## 4.4 Spoken Question Answering

### 4.4 口语问答

**Metrics.** To evaluate the speech intelligence capabilities of our model, we follow previous work (Défossez et al., 2024) and utilize three standard spoken question answering benchmarks: LlamaQ, WebQ, and TriviaQA. We report the accuracy (Acc) under both speech-to-text ( $S \rightarrow T$ ) and speech-to-speech ( $S \rightarrow S$ ) settings. For speech-to-speech setting, we leverage Whisper-large-v3 (Radford et al., 2023) to obtain the transcription of generated speech. Additionally, we report the takeover rate (TOR) across three benchmarks to quantify the frequency of model responses, serving as an indicator of the model’s turn-taking behavior.

**指标。** 为了评估我们模型的语音智能能力，我们遵循先前的工作 (Défossez 等人, 2024)，并使用三个标准的语音问答基准测试：LlamaQ、WebQ 和 TriviaQA。我们在语音到文本 ( $S \rightarrow T$ ) 和语音到语音 ( $S \rightarrow S$ ) 两种设置下报告准确率 (Acc)。对于语音到语音的设置，我们利用 Whisper-large-v3 (Radford 等人, 2023) 来获取生成语音的转录。此外，我们在三个基准测试中报告接管率 (TOR)，以量

化模型响应的频率，作为模型轮流行为的指标。

Result. As presented in Table 1, Lychee-FD demonstrates superior spoken question answering capabilities. It achieves the highest average accuracy across both speech-to-text ( $S \rightarrow T$ ) and speech-to-speech ( $S \rightarrow S$ ) settings. Compared to the previous SOTA native full-duplex model, Fun-Audio-Chat, Lychee-FD delivers a substantial improvement of 7.4% in  $S \rightarrow S$  accuracy and 8.8% in  $S \rightarrow T$  accuracy. Even when compared to system-level pipelines like VITA-1.5, our end-to-end approach demonstrates superior reasoning capabilities (10.8% in  $S \rightarrow S$  and 0.7% in  $S \rightarrow T$ ), validating the effectiveness of our framework. Furthermore, Lychee-FD maintains a perfect TOR of 100%, confirming that this exceptional knowledge retention is achieved without compromising seamless, real-time interaction stability.

结果。如表 1 所示，荔枝-FD 在口语问答任务中表现出色。它在语音到文本 ( $S \rightarrow T$ ) 和语音到语音 ( $S \rightarrow S$ ) 设置下均实现了最高的平均准确率。与之前最先进的原生全双工模型 Fun-Audio-Chat 相比，荔枝-FD 在  $S \rightarrow S$  准确率上提升了 7.4%，在  $S \rightarrow T$  准确率上提升了 8.8%。即使与系统级流程如 VITA-1.5 进行比较，我们的端到端方法也展示了更强的推理能力（在  $S \rightarrow S$  上为 10.8%，在  $S \rightarrow T$  上为 0.7%），验证了我们框架的有效性。此外，荔枝-FD 保持了 100% 的 TOR，确认了这种卓越的知识保留是在不影响无缝、实时交互稳定性的前提下实现的。

Ablation. To systematically investigate the individual contributions of our two proposed innovations, we conducted ablation studies on two variants: w/o Sem-Channel, which replaces the semantic alignment channel

消融实验。为了系统地研究我们提出的两个创新方法各自的贡献，我们在两个变体上进行了消融实验：w/o Sem-Channel，即用语义对齐通道替换。

| ModelFDBench     |      |      |      |      |      |      | FullDuplexBench 1.0 |        |       |       |       | FullDuplexBench 1.5 |      |      |      |      |
|------------------|------|------|------|------|------|------|---------------------|--------|-------|-------|-------|---------------------|------|------|------|------|
| SRR              | SIR  | EIR  | SRIR | FS   | ED   | IRD  | I-TOR               | B-Freq | B-TOR | T-TOR | P-TOR | Stop                | IRR  | BRR  | Stop | Lat. |
| dGSLM            | -    | -    | -    | -    | -    | -    | -                   | 91.7   | 1.5   | 69.1  | 97.5  | 93.5                | 2523 | -    | -    | -    |
| Freeze           | 12.9 | 57.2 | 25.7 | 29.5 | 667  | 5413 | 77.5                | 0.1    | 63.6  | 33.6  | 46.3  | 1380                | 27.0 | 63.0 | 660  | 2066 |
| Omni             |      |      |      |      |      |      |                     |        |       |       |       |                     |      |      |      |      |
| VITA 1.5         | 21.0 | 46.1 | 16.3 | 78.3 | 3036 | 9925 | 99.5                | 2.5    | 81.8  | 58.8  | 88.8  | 1523                | 6.0  | 38.0 | 1222 | 2140 |
| FLM-audio        | 7.5  | 69.4 | 1.0  | 0.9  | 989  | 3408 | 91.0                | 0.3    | 61.8  | 96.6  | 56.5  | 4579                | 10.0 | 43.0 | 2439 | 983  |
| Moshi            | 41.4 | 78.8 | 22.1 | 73.9 | 1895 | 1421 | 87.5                | 5.1    | 36.4  | 76.4  | 54.1  | 885                 | 61.0 | 26.0 | 1071 | 3034 |
| Lychee-FD (Ours) | 46.3 | 99.7 | 0.4  | 95.8 | 637  | 1210 | 94.5                | 14.6   | 23.4  | 98.3  | 10.0  | 840                 | 78.0 | 69.0 | 570  | 826  |

| 模型       | FDBench |      |      |      |      |      | FullDuplexBench 1.0 |       |        |       |       | FullDuplexBench 1.5 |      |      |      |      |      |
|----------|---------|------|------|------|------|------|---------------------|-------|--------|-------|-------|---------------------|------|------|------|------|------|
|          | SRR     | SIR  | EIR  | SRIR | FS   | ED   | IRD                 | I-TOR | B-Freq | B-TOR | T-TOR | P-TOR               | 停止   | IRR  | BRR  | 停止   | Lat. |
| dGSLM    | -       | -    | -    | -    | -    | -    | -                   | 91.7  | 1.5    | 69.1  | 97.5  | 93.5                | 2523 | -    | -    | -    | -    |
| Freeze   | 12.9    | 57.2 | 25.7 | 29.5 | 667  | 5413 | 77.5                | 0.1   | 63.6   | 33.6  | 46.3  | 1380                | 27.0 | 63.0 | 660  | 2066 |      |
| Omni     |         |      |      |      |      |      |                     |       |        |       |       |                     |      |      |      |      |      |
| VITA 1.5 | 21.0    | 46.1 | 16.3 | 78.3 | 3036 | 9925 | 99.5                | 2.5   | 81.8   | 58.8  | 88.8  | 1523                | 6.0  | 38.0 | 1222 | 2140 |      |

|                                |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|--------------------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| FLM-<br>audio                  | 7.5  | 69.4 | 1.0  | 0.9  | 989  | 3408 | 91.0 | 0.3  | 61.8 | 96.6 | 56.5 | 4579 | 10.0 | 43.0 | 2439 | 983  |
| 摩<br>西                         | 41.4 | 78.8 | 22.1 | 73.9 | 1895 | 1421 | 87.5 | 5.1  | 36.4 | 76.4 | 54.1 | 885  | 61.0 | 26.0 | 1071 | 3034 |
| 荔<br>枝-<br>FD<br>(我<br>们<br>的) | 86.3 | 99.7 | 0.4  | 95.8 | 637  | 1210 | 94.5 | 14.6 | 23.4 | 98.3 | 10.0 | 840  | 78.0 | 69.0 | 570  | 826  |

Table 2: Performance comparison of full-duplex interaction capabilities and efficiency. Despite interaction behavior metrics, we also report efficiency metrics of each benchmark (FSED, IRD, Lat. Stop.), measured in milliseconds (ms).  $\uparrow$  indicates higher is better. Bold denotes best results and underlined denotes second best.

表 2: 全双工交互能力与效率的性能对比。尽管交互行为指标, 我们还报告了每个基准测试的效率指标 (FSED、IRD、Lat. Stop.), 以毫秒 (ms) 为单位。 $\uparrow$  表示数值越高越好。加粗表示最佳结果, 下划线表示次佳结果。

with sparse time-aligned text, and w/o Param-Sep, which removes the hierarchical parameter separation to employ a fully shared architecture. As shown in Table 1, when applying the time-aligned text, we observe a significant performance decline in both  $S \rightarrow T$  (5.6%) and  $S \rightarrow S$  (5.4%) settings. This parallel degradation confirms our hypothesis regarding semantic dilution: the sparse supervision provided by time-aligned text fails to sustain robust linguistic modeling, which in turn causes knowledge degradation. In contrast, the w/o Param-Sep setting reveals severe modality interference. While its text generation capability remains relatively stable, its speech accuracy suffers a catastrophic drop to 27.6%. This disparity validates our optimization dynamics analysis (as illustrated in Figure 2): without physically disentangling the shared parameters, the optimization landscape becomes dominated by semantic modeling, effectively suppressing the learning of acoustic features in deep layers.

带有稀疏时间对齐文本, 并且不使用参数分离 (w/o Param-Sep), 这消除了层级参数分离, 采用完全共享的架构。如表 1 所示, 当应用时间对齐文本时, 我们观察到在  $S \rightarrow T$  (5.6%) 和  $S \rightarrow S$  (5.4%) 设置下性能显著下降。这种并行的性能下降验证了我们关于语义稀释的假设: 时间对齐文本提供的稀疏监督无法维持稳健的语言建模, 从而导致知识退化。相比之下, 不使用参数分离 (w/o Param-Sep) 的设置表现出严重的模态干扰。虽然其文本生成能力相对稳定, 但其语音准确性却急剧下降至 27.6%。这种差异验证了我们的优化动态分析 (如图 2 所示): 在没有物理上解耦共享参数的情况下, 优化景观被语义建模所主导, 从而有效地抑制了深层网络中声学特征的学习。

Discussion. We highlight two critical phenomena that distinguish Lychee-FD from existing paradigms. First, our model not only recovers the performance of its half-duplex backbone (StepAudio-2-mini) but surpasses it in both  $S \rightarrow T$  (+0.2%) and  $S \rightarrow S$  (+5.3%) settings. This performance gain, achieved without increasing model depth, strongly validates that our framework realizes robust knowledge retention while maintaining the inference efficiency of the native end-to-end model. This underscores the pivotal role of explicit semantic modeling in driving model intelligence, offering valuable insights for future human-machine interaction systems. Second, Lychee-FD achieves the smallest modality gap among all native CDM models (e.g., FLM-Audio and Moshi) and remains competitive with Thinker-Talker architectures, without requiring intricate multi-stage training. This demonstrates that by physically decoupling semantic and acoustic modeling, Lychee-FD effectively resolves modality interference. Consequently, our approach allows the model to articulate nature acoustic responses without corrupting its underlying semantic logic, providing a streamlined solution to the efficiency-intelligence trade-off.

讨论。我们强调了两个关键现象, 这些现象使 Lychee-FD 与现有范式区分开来。首先, 我们的模型不仅恢

复了其半双工骨干网络 (StepAudio-2-mini) 的性能, 而且在  $S \rightarrow T(+0.2\%)$  和  $S \rightarrow S(+5.3\%)$  设置下都超越了它。这种性能提升是在不增加模型深度的情况下实现的, 这强烈验证了我们的框架在实现稳健的知识保留的同时, 保持了原生端到端模型的推理效率。这突显了显式语义建模在推动模型智能中的关键作用, 为未来的人机交互系统提供了宝贵的见解。其次, Lychee-FD 在所有原生 CDM 模型 (例如 FLM-Audio 和 Moshi) 中实现了最小的模态间隙, 并且在不需要复杂的多阶段训练的情况下, 仍然能够与 Thinker-Talker 架构保持竞争力。这表明, 通过物理上解耦语义和声学建模, Lychee-FD 有效地解决了模态干扰问题。因此, 我们的方法使得模型能够在不破坏其底层语义逻辑的情况下, 表达自然的声学响应, 为效率与智能之间的权衡提供了一个简洁的解决方案。

## 4.5 Full-duplex Chatting

### 4.5 全双工聊天

**Metrics.** For the Full-duplex Chatting, we select three mainstream benchmarks: FDBench (Peng et al., 2025), FullDuplexBench 1.0, and FullDuplexBench 1.5 (Lin et al., 2025a). We follow the recommended settings of these benchmarks for evaluating both baselines and our method. Specifically, FDBench assesses turn-taking and interruption behaviors using Success-Replies Rate (SRR), Success-Interrupts Rate (SIR), Early-Interrupts Rate (EIR), and Success-Replies-to-Interrupts Rate (SRIR), alongside timing metrics such as First-Speech-Emit-Delay (FSED) and Interrupt-Response-Delay (IRD). FullDuplexBench 1.0 consists of four subsets: interruption (I), assistant backchannel (B), turn-taking (T), and user pause (P). In addition to the takeover rate (TOR) and Interruption stop delay (Stop.), we also report the frequency of generated backchannels during user speech (B-Freq). Finally, FullDuplexBench 1.5 utilizes GPT-4o-1124 to classify model responses to user interruptions and backchannels into four categories (Response, Resume, Uncertain, or Unknown), reporting the Interruption-Response Rate (IRR), Backchannel-Resume Rate (BRR), as well as the interruption stop delay and response latency.

**指标.** 对于全双工聊天, 我们选择了三个主流基准测试: FDBench (Peng 等, 2025)、FullDuplexBench 1.0 和 FullDuplexBench 1.5 (Lin 等, 2025a)。我们在评估基线方法和我们的方法时, 遵循这些基准测试推荐的设置。具体来说, FDBench 使用成功回复率 (SRR)、成功中断率 (SIR)、早期中断率 (EIR) 和成功回复中断率 (SRIR) 来评估轮流发言和中断行为, 同时结合诸如首次发言延迟 (FSED) 和中断响应延迟 (IRD) 等时间指标。FullDuplexBench 1.0 包含四个子集: 中断 (I)、助手回传 (B)、轮流发言 (T) 和用户停顿 (P)。除了接管率 (TOR) 和中断停止延迟 (Stop) 外, 我们还报告了用户发言期间生成的回传频率 (B-Freq)。最后, FullDuplexBench 1.5 使用 GPT-4o-1124 将模型对用户中断和回传的响应分类为四个类别 (响应、继续、不确定或未知), 并报告中断响应率 (IRR)、回传继续率 (BRR) 以及中断停止延迟和响应延迟。

**Result.** As presented in Table 2, Lychee-FD achieves state-of-the-art performance across 10 of the 11 evaluated interaction metrics, demonstrating superior capability in managing complex conversational dynamics, including interruption, backchanneling, dynamic turn-taking, and pause handling. While VITA-1.5 exhibits a marginally higher I-TOR on FullDuplexBench 1.0, its consistently high B-TOR and P-TOR reveal a tendency towards aggressive, indiscriminate speech rather than intelligent turn-taking. In contrast, Lychee-FD maintains a balanced interaction profile, effectively distinguishing between user pauses and interruptions. Notably, on the challenging FullDuplexBench 1.5, our model delivers a substantial 28.5% average improvement over system-level baselines (e.g., Freeze-Omni) that rely on external VAD. Crucially, this empirical breakthrough directly validates our core scientific insight: by resolving the inherent gradient conflicts that plague fully shared

**结果.** 如表 2 所示, Lychee-FD 在 11 个评估的交互指标中的 10 个上实现了最先进的性能, 展示了其在管理复杂对话动态方面的卓越能力, 包括中断、反馈、动态回合转换和停顿处理。虽然 VITA-1.5 在 FullDuplexBench 1.0 上表现出略高的 I-TOR, 但其持续较高的 B-TOR 和 P-TOR 表明其倾向于非智能的、随意的发言, 而非智能的回合转换。相比之下, Lychee-FD 保持了平衡的交互特征, 能够有效区分用户的停顿和中断。值得注意的是, 在具有挑战性的 FullDuplexBench 1.5 上, 我们的模型在系统级基线 (如依赖外部 VAD 的 Freeze-Omni) 上实现了显著的 28.5% 平均提升。关键的是, 这一实证突破直接验证了我们的核心科学洞察: 通过解决完全共享模型中固有的梯度冲突问题,

architectures, Lychee-FD preserves robust semantic awareness during simultaneous listening and speaking. Consequently, this result not only proves that our model realizes truly natural, fluid, and immersive interaction, but also confirms that LLMs can intrinsically function as highly effective dialog managers. By eliminating handcrafted signal processing modules, our approach provides an efficient, practical solution to the core problem of full-duplex communication, substantially pushing forward the frontier of native end-to-end SLMs.

架构方面，Lychee-FD 在同时进行听和说的过程中保持了强大的语义感知能力。因此，这一结果不仅证明了我们的模型能够实现真正自然、流畅和沉浸式的交互，还证实了大语言模型（LLMs）可以内在地作为高效的对话管理器发挥作用。通过消除手工设计的信号处理模块，我们的方法为全双工通信的核心问题提供了一个高效且实用的解决方案，显著推动了原生端到端语音语言模型（SLMs）的发展前沿。

**Latency.** We evaluate inference efficiency through two categories: first response latency (FSED, Lat.) and interruption response latency (IRD, Stop.). Regarding the former, Lychee-FD achieves the lowest latency across all benchmarks. Since our hierarchical parameter separation strategy introduces no additional model depth, we can leverage standard pipeline parallelism for speedup, demonstrating the efficiency of our architectures. Further details about this real-time parallel inference are provided in Appendix 10. For Interruption Response Latency, our model demonstrates an even greater advantage, achieving the lowest stop latency when meeting an interruption (e.g., 570ms Stop. on FullDuplexBench 1.5). It is worth noting that interruption latency is a function of both model processing speed and interaction accuracy—since a model must first correctly identify an interruption before it can stop generating. The superior performance confirms that our architecture incurs no computational overhead, and that its robust semantic awareness enables rapid, accurate reactions to user interventions.

延迟。我们通过两个类别来评估推理效率：首次响应延迟（FSED, Lat.）和中断响应延迟（IRD, Stop.）。关于前者，Lychee-FD 在所有基准测试中实现了最低的延迟。由于我们的分层参数分离策略不会引入额外的模型深度，因此我们可以利用标准的流水线并行技术来提升速度，这展示了我们架构的高效性。有关这种实时并行推理的更多细节，请参见附录 10。对于中断响应延迟，我们的模型展现出更大的优势，在遇到中断时实现了最低的停止延迟（例如，在 FullDuplexBench 1.5 上为 570ms Stop.）。值得注意的是，中断延迟取决于模型处理速度和交互准确性——因为模型必须首先正确识别中断，才能停止生成。卓越的性能确认了我们的架构不会产生计算开销，并且其强大的语义感知能力使其能够快速、准确地响应用户的干预。

## 4.6 Speech Generation

### 4.6 语音生成

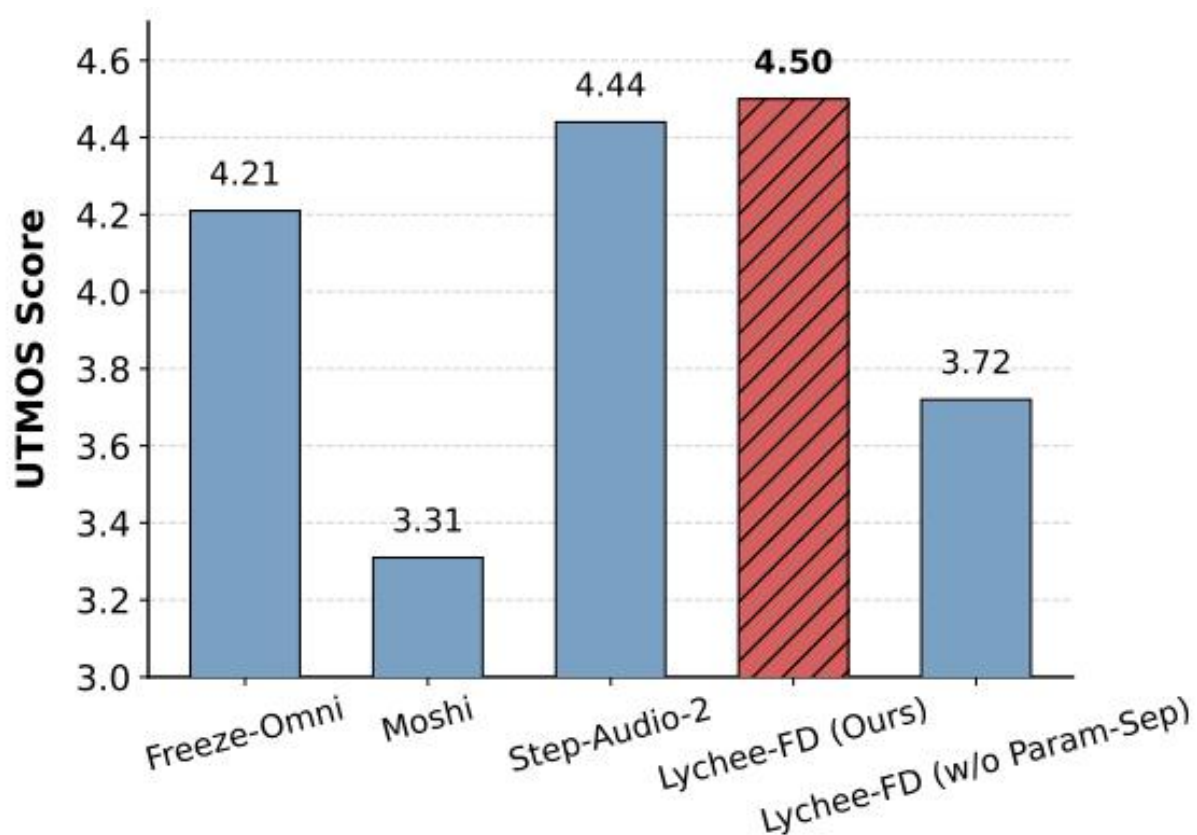


Figure 5: images/962959ecc7cdc6239e95c64407b9e9adea8b8de852018465ae6acf04e820c1e0.jpg

Figure 4: Comparison of speech synthesis quality via UTMOS. Lychee-FD (w/o Param-Sep) denotes the variant without hierarchical parameter separation strategy.

图 4: 通过 UT MOS 比较语音合成质量。Lychee-FD (无 Param-Sep) 表示不采用分层参数分离策略的变体。

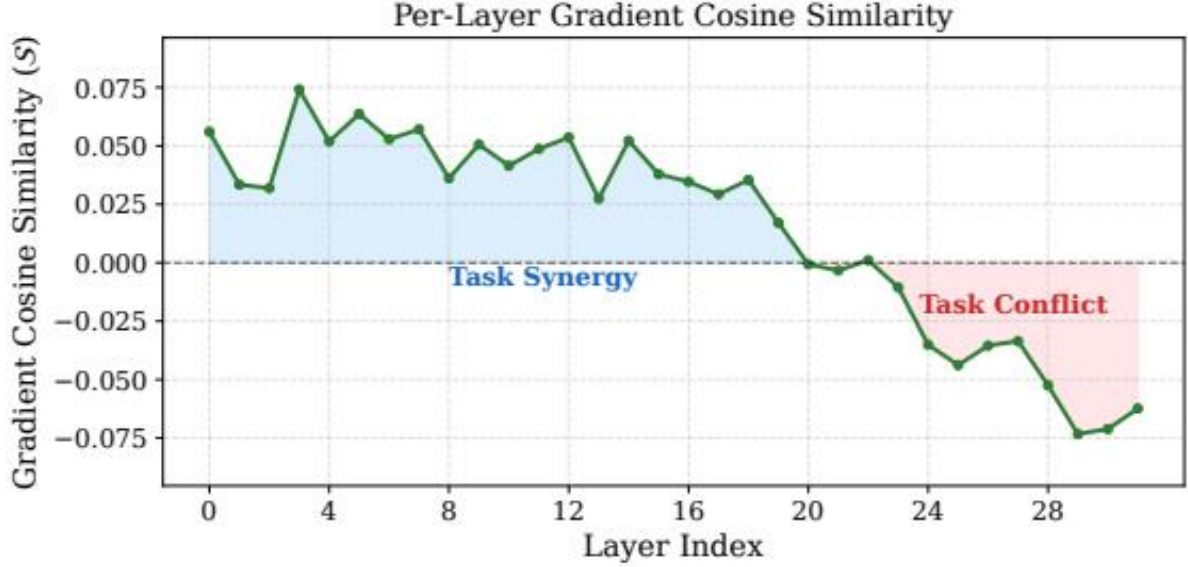


Figure 6: images/d6786a166389f2ac1475fe2e389c0c3d6c2554cb4a619156952a21ff8646077d.jpg

Figure 5: Gradient Cosine Similarity on the Moshi architecture. The transition from task synergy in shallow layers to task conflict in deep layers confirms that modality interference is a universal phenomenon in fully shared end-to-end SLMs.

图 5：在 Moshi 架构上的梯度余弦相似度。从浅层任务协同向深层任务冲突的转变证实了模态干扰是全共享端到端 SLMs 中普遍存在的现象。

**Metrics.** To investigate the quality of the generated speech, we evaluate the content consistency and speech naturalness of the model on LlamaQ dataset. Specifically, we report the Word Error Rate (WER) between the generated text and the transcribed speech to measure content consistency. Additionally, we employ UTMOS (Saeki et al., 2022), a trained speech quality assessment model, to score the naturalness of the generated audio.

指标。为了调查生成语音的质量，我们在 LlamaQ 数据集上评估模型的内容一致性与语音自然度。具体而言，我们报告生成文本与转录语音之间的单词错误率（WER），以衡量内容一致性。此外，我们采用 UTMOS（Saeki 等，2022）这一训练好的语音质量评估模型，对生成的音频的自然度进行评分。

**Result.** High-fidelity speech synthesis serves as the cornerstone of voice interaction, significantly elevating the immersive quality of full-duplex conversations. As illustrated in Figure 4, Lychee-FD achieves the highest UTMOS score of 4.50, surpassing Freeze-Omni, Moshi and even its half-duplex backbone. This result empirically validates that our proposed hierarchical parameter separation strategy effectively prevents acoustic modeling from semantic interference, thereby preserving fine-grained prosodic details that are often lost in shared architectures. When the parameter separation is removed, we observe a significant decline in speech quality. This finding corroborates our hypothesis regarding modality interference from an acoustic perspective, demonstrating that forcing the acoustic head to share deep layers with text processing objectives inevitably degrades audio fidelity. Ultimately, by physically disentangling the modeling pathways, Lychee-FD not only improves generation accuracy but also synthesizes speech with richer acoustic details, delivering a more natural and enjoyable conversational experience.

结果。高质量的语音合成是语音交互的基石，显著提升了全双工对话的沉浸感。如图 4 所示，Lychee-FD 实现了 4.50 的 UTMOS 得分，超过了 Freeze-Omni、Moshi，甚至超过了其半双工骨干模型。这一结果实证验证了我们提出的层次化参数分离策略能够有效防止声学建模受到语义干扰，从而保留通常在共享架构中丢失的细粒度韵律细节。当去除参数分离时，我们观察到语音质量显著下降。这一发现从声学角度证实了我们关于模态干扰的假设，表明强制声学头与文本处理目标共享深层次不可避免地会降低音频保真



度。最终，通过物理上解耦建模路径，Lychee-FD 不仅提高了生成的准确性，还合成出更具丰富声学细节的语音，带来了更加自然和愉悦的对话体验。

## 5 Optimization Dynamics Analysis Extension

### 5 优化动态分析扩展

To verify the universality of our findings, we extend the gradient cosine similarity analysis to Moshi (Défossez et al., 2024), investigating whether deep-layer gradient conflicts are architecture-specific artifacts or a fundamental bottleneck across different FDSLMS. Following our established methodology, we find that Moshi’s optimization dynamics exhibit a strikingly similar trend (Figure 5). Shallow layers (0-19) maintain positive similarity ( $\mathcal{S}^{(l)} > 0$ ), indicating task synergy. Conversely, as depth increases, the similarity sharply declines to distinctly negative values in deep layers (23-31). This highlights the optimization divergence between text and speech tasks. This cross-architecture validation empirically confirms that inherent gradient conflicts between semantic and acoustic modeling are not isolated anomalies, but a universal bottleneck in fully shared paradigms. Consequently, it firmly substantiates the necessity and generalizability of our proposed Hierarchical Parameter Separation.

为了验证我们发现的普适性，我们将梯度余弦相似性分析扩展到 Moshi (Défossez 等, 2024)，探讨深层梯度冲突是特定于架构的伪现象，还是在不同 FDSLMS 中普遍存在的根本性瓶颈。遵循我们已建立的方法，我们发现 Moshi 的优化动态表现出非常相似的趋势（图 5）。浅层（0-19 层）保持正相似度（ $\mathcal{S}^{(l)} > 0$ ），表明任务协同。相反，随着深度增加，相似度在深层（23-31 层）急剧下降至明显负值。这突显了文本与语音任务之间的优化分歧。这种跨架构的验证实证地确认了语义建模与声学建模之间的固有梯度冲突并非孤立的异常现象，而是全共享范式中的普遍瓶颈。因此，这有力地证明了我们提出的分层参数分离方法的必要性和普适性。

## 6 Conclusion

### 6 结论

In this paper, we address a critical challenge and a long-standing goal for the speech and NLP communities: developing seamless, high-performance native full-duplex spoken language models. To the best of our knowledge, we are the first to uncover the fundamental root cause of modality interference. Through an in-depth analysis of model optimization dynamics, we reveal that the inability to simultaneously listen and speak stems from inherent gradient conflicts within a shared deep parameter space, which specifically manifest as optimization divergence and semantic dilution. Inspired by these observations, we introduce Lychee-FD. Our framework elegantly resolves these bottlenecks by decoupling conflicting modalities via a hierarchical parameter separation strategy, coupled with a semantic alignment channel to enforce robust knowledge retention. Extensive experiments demonstrate that Lychee-FD significantly advances the state-of-the-art performance, successfully reconciling ultra-low latency interaction with deep speech intelligence. Ultimately, our work substantially pushes forward the frontier of full-duplex SLMs research, paving the way for the next generation of natural, fluid, and immersive human-machine interaction.

在本文中，我们探讨了语音和自然语言处理（NLP）社区面临的一个关键挑战和长期目标：开发无缝、高性能的原生全双工语音语言模型。据我们所知，我们是首个揭示模态干扰根本原因的团队。通过深入分析模型优化动态，我们发现无法同时听和说的原因在于共享的深层参数空间中固有的梯度冲突，这具体表现为优化发散和语义稀释。受这些观察启发，我们引入了 Lychee-FD。我们的框架通过分层参数分离策略将冲突的模态解耦，并结合语义对齐通道以强制保留稳健的知识。大量实验表明，Lychee-FD 显著提升了当前最先进的性能，成功地将超低延迟交互与深度语音智能相结合。最终，我们的工作大大推动了全双工语言模型（SLMs）研究的前沿，为下一代自然、流畅和沉浸式的人机交互铺平了道路。

## Limitations

### 限制

While Lychee-FD enables the development of a high-performance native full-duplex framework with highly responsive acoustic processing capabilities, deploying such systems in unconstrained, real-world open-mic scenarios remains a promising yet challenging next research frontier. At present, our model implements highly sensitive interruption detection and can effectively suspend speech output upon any user barge-in. However, discriminating between intentional user instructions and incidental background side-talk (as illustrated in Appendix 8.2) remains a non-trivial challenge to address. This limitation is tentatively attributed primarily to the current data synthesis paradigm, which is largely centered on direct two-party interactions and lacks detailed intent annotations for complex multi-speaker scenarios. This issue points to a critical gap in data coverage within the field, rather than stemming from an architectural bottleneck of the hierarchical framework we propose. Moving forward, we hope our preliminary work can provide a modest impetus for subsequent research in the community. By constructing diverse open-mic datasets and integrating prosodic features for intent disambiguation, future studies may build upon this initial framework to develop truly context-aware and adaptive selective interruption mechanisms for next-generation SLMs.

虽然荔枝-FD 使得开发一个高性能的原生全双工框架成为可能，该框架具有高度响应的声学处理能力，但在不受限制的真实世界开放麦克风场景中部署此类系统仍然是一个充满前景但具有挑战性的研究前沿。目前，我们的模型实现了高度敏感的中断检测，并且在用户插话时可以有效地暂停语音输出。然而，区分有意的用户指令和偶然的背景旁白（如附录 8.2 所示）仍然是一个非同寻常的挑战。这一限制目前主要归因于当前的数据合成范式，该范式主要围绕直接的双人交互展开，并缺乏对复杂多说话人场景的详细意图标注。这一问题反映了该领域数据覆盖的关键缺口，而非我们提出的分层框架的架构瓶颈。未来，我们希望我们的初步工作能够为社区后续的研究提供一定的推动力。通过构建多样化的开放麦克风数据集，并整合语调特征以澄清意图，未来的研究可以在此初始框架的基础上发展出真正具有上下文感知和自适应选择性中断机制的下一代 SLMs。